

Limits at the Edges

Answer Key

AP Calculus AB/BC

Quick Answers

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| 1. (a) left-hand limit = -1 , (b) right-hand limit = 1 | 6. -2 |
| 2. 0 | 7. (a) the limit at $x = 2 = -4$, (b) the limit as x approaches 3 from the right = ∞ |
| 3. $\frac{3}{4}$ | 8. 0 |
| 4. (a) from the left = $-\infty$, (b) from the right = ∞ | 9. -4 |
| 5. (a) left-hand limit = 5 , (b) right-hand limit = 8 , — | 10. (a) as x approaches positive infinity = 1 , (b) as x approaches negative infinity = -1 |
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What is verified

15 of 16 answers machine-checked against SymPy, across 10 problems.
— marks an answer only you can judge — problem 5. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus AB/BC

LIM-1 – problems 1, 2, 3, 4, 6, 7, 8, 9, 10

LIM-2 – problem 5

Difficulty 1–4 of 5 across 16 tagged checks.

Common wrong answers

8. If they got 2.5: divided the leading coefficients without comparing the degrees

1. $f(x) = \frac{|x-3|}{x-3}$. Find (a) $\lim_{x \rightarrow 3^-} f(x)$ and (b) $\lim_{x \rightarrow 3^+} f(x)$.

Solution: Split the absolute value by cases. For $x < 3$ we have $x - 3 < 0$, so $|x - 3| = -(x - 3)$ and $f(x) = \frac{-(x-3)}{x-3} = -1$ on that whole side. The function is the constant -1 there, so the left-hand limit is -1

For $x > 3$ we have $x - 3 > 0$, so $|x - 3| = x - 3$ and $f(x) = 1$ on that whole side, giving 1

Because the two one-sided values differ, $\lim_{x \rightarrow 3} f(x)$ does not exist — this is the standard jump at a sign change.

2. $g(x) = \frac{x-4}{\sqrt{x-4}}$, defined only for $x > 4$. Find $\lim_{x \rightarrow 4^+} g(x)$.

Solution: Write $x-4 = (\sqrt{x-4})^2$, so for $x > 4$

$$g(x) = \frac{(\sqrt{x-4})^2}{\sqrt{x-4}} = \sqrt{x-4}.$$

As $x \rightarrow 4^+$ the quantity $x-4 \rightarrow 0^+$ and its square root does too, so the limit is 0

Only the right-hand limit is asked because g has no left side: $\sqrt{x-4}$ is undefined for $x < 4$.

3. Evaluate $\lim_{x \rightarrow \infty} \frac{3x^2 - 5x + 2}{4x^2 + 7}$.

Solution: Divide every term by the highest power in the denominator, x^2 :

$$\frac{3x^2 - 5x + 2}{4x^2 + 7} = \frac{3 - \frac{5}{x} + \frac{2}{x^2}}{4 + \frac{7}{x^2}}.$$

Each $\frac{c}{x^n} \rightarrow 0$ as $x \rightarrow \infty$, leaving $\frac{3}{4}$. $\frac{3}{4}$

Equivalently: equal degrees, so the limit is the ratio of the leading coefficients. The constants 2 and 7 never enter it.

4. $h(x) = \frac{x+3}{x-5}$. Find (a) $\lim_{x \rightarrow 5^-} h(x)$ and (b) $\lim_{x \rightarrow 5^+} h(x)$.

Solution: At $x = 5$ the numerator is $8 \neq 0$ and the denominator is 0, so $x = 5$ is a vertical asymptote and the answer is decided entirely by the sign of $x - 5$.

For x slightly below 5: numerator near +8, denominator a small *negative*, so the quotient is a large negative number. $-\infty$

For x slightly above 5: numerator near +8, denominator a small *positive*, so the quotient is a large positive number. $+\infty$

5. Piecewise shipping cost $f(x) = 3x - 1$ for $x < 2$ and $f(x) = x^2 + 4$ for $x \geq 2$. Find (a) the left-hand limit at 2, (b) the right-hand limit at 2, (c) whether $\lim_{x \rightarrow 2} f(x)$ exists.

Solution (a): Approaching 2 from below uses only the rule for $x < 2$, so evaluate the limit of $3x - 1$ at 2: $3(2) - 1$. 5

Solution (b): Approaching 2 from above uses only the rule for $x \geq 2$, so evaluate the limit of $x^2 + 4$ at 2: $2^2 + 4$. 8

Solution (c) — instructor-judged, no machine check. The correct conclusion is that $\lim_{x \rightarrow 2} f(x)$ **does not exist**, and the required reason is that the two one-sided limits disagree (5 from the left, 8 from the right), so there is no single number the outputs approach

from both sides — the cost jumps by \$3 at a weight of 2 kg. Award no credit for a bare “DNE” with no reason. Do not accept “the limit is 8 because $f(2) = 8$ ”: the value of the function at the point is not what a limit asks about.

6. Evaluate $\lim_{x \rightarrow -\infty} \frac{2x^3 + x}{5 - x^3}$.

Solution: Divide through by x^3 :

$$\frac{2x^3 + x}{5 - x^3} = \frac{2 + \frac{1}{x^2}}{\frac{5}{x^3} - 1} \longrightarrow \frac{2 + 0}{0 - 1}.$$

The direction of approach does not matter here — every discarded term tends to 0 at both ends — so the limit is -2

7. $k(x) = \frac{x^2 - 4}{x^2 - 5x + 6}$. Find (a) $\lim_{x \rightarrow 2} k(x)$ and (b) $\lim_{x \rightarrow 3^+} k(x)$.

Solution: Factor both parts first — that is what separates the hole from the asymptote:

$$k(x) = \frac{(x - 2)(x + 2)}{(x - 2)(x - 3)} = \frac{x + 2}{x - 3} \quad (x \neq 2).$$

(a) The factor $x - 2$ cancels, so $x = 2$ is a removable discontinuity, a hole. Substitute into the reduced form: $\frac{2+2}{2-3} = \frac{4}{-1}$. -4

(b) The factor $x - 3$ does *not* cancel, so $x = 3$ is a genuine vertical asymptote. As $x \rightarrow 3^+$ the numerator tends to $5 > 0$ and $x - 3$ is a small positive, so the quotient grows without bound. $+\infty$

8. Evaluate $\lim_{x \rightarrow \infty} \frac{5x^2 - 3}{2x^3 + x}$.

Solution: The denominator has the higher degree, so divide by x^3 :

$$\frac{5x^2 - 3}{2x^3 + x} = \frac{\frac{5}{x} - \frac{3}{x^3}}{2 + \frac{1}{x^2}} \longrightarrow \frac{0 - 0}{2 + 0}.$$

The bottom outgrows the top, so the quotient is squeezed to 0

Comparing leading coefficients only ($\frac{5}{2}$) is the classic error: that shortcut is valid only when the degrees are equal.

9. Evaluate $\lim_{x \rightarrow 2^-} \frac{x^2 - 4}{|x - 2|}$.

Solution: Approaching from the left means $x < 2$, so $x - 2 < 0$ and $|x - 2| = -(x - 2) = 2 - x$. Factor the numerator and cancel:

$$\frac{x^2 - 4}{|x - 2|} = \frac{(x - 2)(x + 2)}{-(x - 2)} = -(x + 2) \quad (x < 2).$$

Now let $x \rightarrow 2^-$ in $-(x + 2)$: the value tends to $-(2 + 2)$.

-4

(From the right the same work gives $+4$, so the two-sided limit does not exist.)

10. Challenge. $r(x) = \frac{\sqrt{9x^2 + 4}}{3x - 1}$. Find (a) $\lim_{x \rightarrow \infty} r(x)$ and (b) $\lim_{x \rightarrow -\infty} r(x)$.

Solution: The key identity is $\sqrt{x^2} = |x|$, not x . Divide numerator and denominator by $|x|$, noting $\sqrt{9x^2 + 4}/|x| = \sqrt{9 + 4/x^2}$.

(a) For $x \rightarrow \infty$ we have $|x| = x$, so

$$r(x) = \frac{\sqrt{9 + \frac{4}{x^2}}}{3 - \frac{1}{x}} \longrightarrow \frac{\sqrt{9}}{3} = \frac{3}{3}. \quad \boxed{1}$$

(b) For $x \rightarrow -\infty$ we have $|x| = -x$, so dividing the denominator by $|x|$ introduces a minus sign:

$$r(x) = \frac{\sqrt{9 + \frac{4}{x^2}}}{-3 + \frac{1}{x}} \longrightarrow \frac{3}{-3}. \quad \boxed{-1}$$

A numerator built from a square root is never negative, while $3x - 1$ is very negative at the left end — so a negative answer is exactly what the graph predicts.